

ORIGINAL ARTICLE

Sleep Apnea–Specific Hypoxic Burden, Symptom Subtypes, and Risk of Cardiovascular Events and All-Cause Mortality

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Abstract

Rationale: Data from population-based cohorts suggest that symptom subtypes and obstructive sleep apnea (OSA)–specific hypoxic burden (HB) could help to better identify patients with OSA at high cardiovascular (CV) risk.

Objectives: We aimed to evaluate whether those new markers are associated with the risk of major adverse CV events (MACE) in clinical setting.

Methods: Data from the Pays de la Loire cohort were linked to health administrative data to identify the occurrence of MACE (a composite outcome including all-cause mortality, acute myocardial infarction, stroke, and unplanned coronary revascularization) in patients with newly diagnosed OSA and no overt CV disease. Latent class analysis was used to identify subtypes based on eight clinically relevant variables. HB was defined as the total area under the respiratory event–related desaturation curve. Cox proportional hazards models were used to evaluate the association of symptom subtypes and HB with MACE.

Measurements and Main Results: Four symptom subtypes were identified (minimally symptomatic [22.0%], disturbed sleep [17.5%], excessively sleepy [49.8%], and moderately sleepy [10.6%]). After a median follow-up of 78 months (interquartile range, 52–109), 592 (11.05%) of 5,358 patients experienced MACE. In a fully adjusted model, HB and overall nocturnal hypoxemia assessed by sleep time with oxygen saturation <90% were the only predictors of MACE (hazard ratio, 1.21; 95% confidence interval, 1.07–1.38; and hazard ratio, 1.34; 95% confidence interval, 1.16–1.55, respectively). The association appeared stronger toward younger patients and women.

Conclusion: In clinical setting, patients with OSA who demonstrate elevated OSA-specific HB are at higher risk of a CV event and all-cause mortality. Symptom subtypes were not associated with MACE after adjustment for confounders.

Keywords: sleep apnea; symptom subtypes; cardiovascular disease; sleepiness; cluster analysis

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This article has a related editorial.

This article has an online supplement, which is accessible from this issue's table of contents at www.atsjournals.org.

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At a Glance Commentary

Scientific Knowledge on the

Subject: Significant efforts were made to identify new processes, based on clinical symptoms and on new markers derived from sleep recordings, to better identify patients at high cardiovascular risk in obstructive sleep apnea (OSA). Those new tools were never evaluated in clinical settings.

What This Study Adds to the

Field: In patients with newly diagnosed OSA, sleep apnea-specific hypoxic burden derived from sleep recordings is an independent predictor of incident cardiovascular events and death that could be used in clinical practice. In contrast to observations from population-based cohorts, clinical subtypes based on OSA symptoms have no independent predictive value in clinical-based samples.

Obstructive sleep apnea (OSA) is a common sleep disorder with an estimated prevalence of 20–30% in men and 10–15% in women (1, 2) that is described as a potential risk factor for the development of cardiovascular (CV) diseases (3). However, the association between OSA and CV diseases and all-cause mortality is not homogeneously found in all published studies and in all subtypes of patients with OSA (4–8). Furthermore, three recent randomized controlled trials failed to demonstrate any significant impact of continuous positive airway pressure (CPAP) on CV prognosis in patients with moderate to severe OSA at high CV risk (9–11), emphasizing the need to better identify at-risk patients.

Therefore, significant efforts were made to better identify patients whose CV prognosis is clearly impacted by OSA, considering comorbidities, specific symptoms, and novel sleep recording-derived OSA severity markers.

Recent *post hoc* analyses of the Impact of Sleep Apnea Syndrome in the Evolution of Acute Coronary Syndrome (ISAAC) study showed that among patients admitted for acute coronary syndrome, OSA was an independent risk factor only in patients with

no previous heart disease (12). More elaborate subtypes of patients based on clinical symptoms at diagnosis were also identified by means of cluster analysis within both clinical (13–16) and population-based samples (17, 18). The association between the identified subtypes (minimally symptomatic, disturbed sleep, excessively sleepy, and moderately sleepy) and incident CV disease was recently investigated in the SHHS (Sleep Heart Health Study). In this large population-based sample, only the “excessively sleepy” subtype was associated with an increased risk of incident CV events compared with subjects without OSA (18). However, whether symptom subtypes predict CV outcomes has not been evaluated in a clinical setting.

There is increasing evidence that common metrics of OSA severity such as apnea–hypopnea index (AHI) and oxygen desaturation index (ODI) do not adequately foresee the risk of CV diseases. New OSA severity markers, including OSA-specific hypoxic burden (HB) and heart rate variability, have been identified as strong predictors of overall CV morbidity and mortality in population-based samples (19–21). We recently reported that pulse rate variability (PRV), a reliable marker of the sympathovagal balance derived from pulse oximetry, predicts incident atrial fibrillation (AF) and stroke in a clinical-based cohort (22, 23).

The relevance of symptom subtypes and new promising biomarkers derived from sleep recordings to better identify patients with OSA at high CV risk needs to be evaluated in a clinical setting. Therefore, we aimed to evaluate whether OSA clinical subtypes, HB, and PRV were associated with a composite outcome of all-cause mortality and CV events in a large multicenter clinic-based cohort of patients with newly diagnosed OSA and no overt CV disease.

Methods

Study Design and Population

This study was conducted on the Pays de la Loire Sleep Cohort, linked with data from the French administrative healthcare database (SNDS) (*see* online supplement for details on the Pays de la Loire Sleep Cohort and the linking process) (22, 24). We included overt CV disease-free patients with OSA with an AHI ≥ 5 events per hour on manually scored polysomnography (PSG) or type 3 home sleep apnea testing (HSAT) (CID102L8DTM

and CID102LTM, respectively, CIDELEC) between May 15, 2007, and December 31, 2019, using the following criteria: apnea was defined as at least a 90% decrease in the oronasal sensor signal, and hypopnea was defined as at least a 30% decrease in the nasal pressure signal combined with either $\geq 3\%$ arterial oxygen desaturation or an arousal (PSG), both lasting at least 10 seconds (25). All patients had given their written informed consent.

HB and Pulse Rate Variability

The HB was calculated as previously described (22). Briefly, HB was defined as the area under the desaturation curve associated with respiratory events. For each individually identified apnea or hypopnea (identified regardless of associated desaturation), the preevent baseline saturation was defined as the maximum oxygen saturation as measured by pulse oximetry during the 100 seconds before the end of the event. The area under this baseline value was calculated over a subject-specific search window for each event, obtained from an ensemble-averaged desaturation.

Pulse rate variability analyses were performed as previously described (22) based on the oximeter's photoplethysmography signal extracted from diagnostic sleep studies. Time domain analysis included the SD of normal-to-normal (NN) beat intervals, and the root mean square of successive NN differences. Frequency domain parameters were extracted from the spectral analysis of the NN intervals. Normalized low-frequency (LF, from 0.04 to 0.15 Hz) and high-frequency (HF, from 0.15 to 0.4 Hz) powers were calculated. The LF-to-HF ratio evaluates the balance between the sympathetic and parasympathetic systems.

Outcome Variables

Major adverse CV events (MACE), the primary composite outcome of the study, was defined using health administrative data as the first hospitalization due to myocardial infarction, stroke, exacerbation of congestive heart failure, revascularization procedure (percutaneous coronary intervention, coronary artery bypass graft surgery), or all-cause death. Participants were followed from their first diagnostic sleep study to the end of December 2019, or the occurrence of a primary outcome, whichever occurred first. The algorithm to identify the first occurrence of CV events or death based on data from the SNDS is provided in the online supplement.

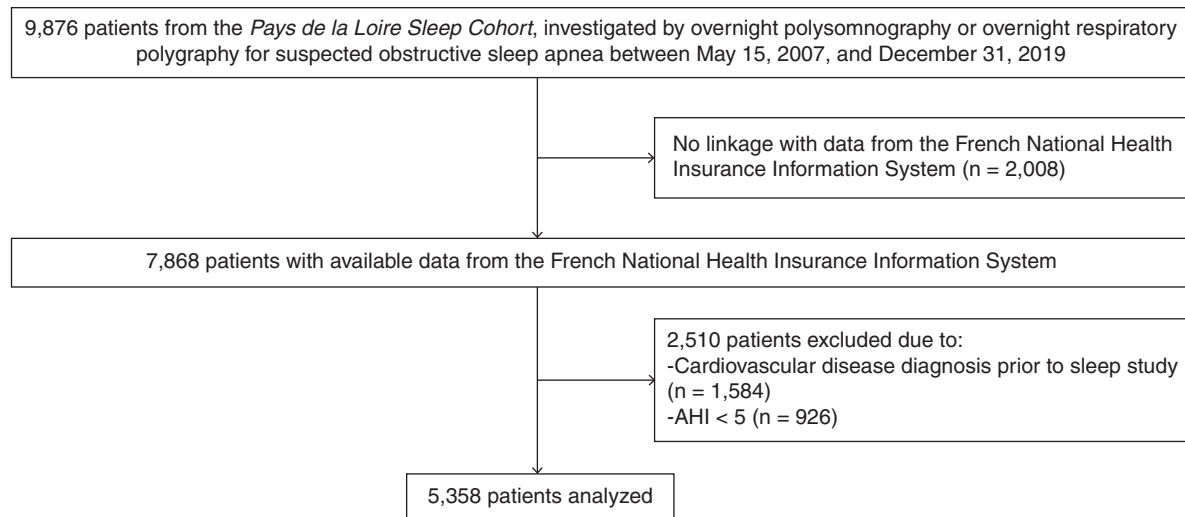


Figure 1. Flow diagram of the study subjects. AHI = apnea–hypopnea index.

Statistical Analysis

Latent class analysis was conducted to identify OSA subtypes as previously described (15, 18). Additional details on the method for OSA subtype analysis are provided in the online supplement. Cox proportional hazards models were used to evaluate the associations between incident MACE with OSA subtypes, AHI, ODI, the percentage of recording time with $\text{SaO}_2 < 90\%$ (T90), PRV, and HB. Associations were evaluated unadjusted and adjusted for covariates. Age, sex, smoking status, body mass index (BMI), the presence of prevalent disease (diabetes, chronic obstructive pulmonary disease, hypertension), type of sleep study (PSG or HSAT), sleep study site, and β -blocker/calcium channel blocker medication were considered as covariates for model 1. A further adjustment of CPAP treatment was proposed in model 2. Further analyses were also performed to consider all-cause mortality and nonfatal CV events

individually. Fine and Gray competing-risk regression was used to account for the competing risk of death for nonfatal CV events risk assessment. Prespecified stratified models were then constructed to examine differential effects of HB on MACE by age, sex, BMI, CPAP treatment status, and type of sleep study (PSG or HSAT). See Table E1 in the online supplement for the definition of covariates and the CPAP treatment follow-up.

All statistical analyses were performed with SAS 9.4 software (SAS Institute). Associations were considered statistically significant for a P value < 0.05 .

Results

Sample Characteristics and Symptom Subtypes

Out of 9,876 patients investigated for suspected OSA between May 15, 2007, and

December 31, 2019, 2,008 were not linked to the SNDS data set, 1,584 had received a diagnosis of CV disease before the sleep study, and 926 had no OSA on sleep recording. The final study sample size included 5,358 patients with OSA without overt CV disease at the time of the diagnostic sleep study (Figure 1).

Clustering analysis identified four optimal clinical subtypes based on symptoms. Figure 2 shows the relative proportion of each symptom across the symptom subtypes. Based on the distribution of observed symptoms and in accordance with previous publications (13, 14, 17, 18), the subtypes were labeled as minimally symptomatic ($n = 1,178$; 22.0%), disturbed sleep ($n = 941$; 17.5%), excessively sleepy ($n = 2,671$; 49.8%), and moderately sleepy ($n = 568$; 10.6%).

Table 1 summarizes the baseline characteristics of these symptom subtypes. The excessively sleepy subtype was

	Cluster 1 Minimally symptomatic ($n = 1,178$)	Cluster 2 Disturbed sleep ($n = 941$)	Cluster 3 Excessively sleepy ($n = 2,671$)	Cluster 4 Moderately sleepy ($n = 568$)	
Complaint of insomnia	6.47	100	9.59	18.96	80%
Loud snoring	47.07	72.68	85.47	100	
Stopped breathing	15.57	55.04	61.69	100	60%
Unrested upon waking	33.77	92	99.05	38.46	
Headache upon waking	13.64	50.08	61.17	8.56	40%
Depressive symptoms	1.52	16.44	33.11	0.72	
Excessive daytime sleepiness	8.31	2.65	82.7	26.75	20%

Figure 2. Prevalence of each variable according to the clusters identified by latent class analysis in 5,358 patients with obstructive sleep apnea. Each colored line represents a variable with prevalence ranging from 0% (yellow) to 100% (red).

Table 1. Baseline Characteristics of the Entire Population by OSA Phenotypes

	Minimally Symptomatic	Disturbed Sleep	Excessively Sleepy	Moderately Sleepy	All	P Value
N	1,178	941	2,671	568	5,358	
Age, yr	63 (52–71)	61 (52–70)	58 (49–67)	64 (55–72)	60 (51–69)	<0.0001
Sex, F, %	32.17	42.40	39.99	17.78	36.34	<0.0001
Body mass index, kg/m ²	30 (26–35)	29 (25–34)	30 (26–35)	29 (26–34)	30 (26–34)	0.0050
Alcohol intake, g/d	0 (0–20)	0 (0–10)	0 (0–10)	10 (0–20)	0 (0–20)	<0.0001
Smoking status, %						0.0004
Never	44.48	47.03	41.83	39.15	43.04	
Current	19.76	21.04	25.01	23.84	23.03	
Former	35.76	31.93	33.16	37.01	33.92	
Prevalent disease, %						
Diabetes	17.19	13.57	12.63	12.31	13.75	0.0007
Hypertension	35.54	32.80	28.65	29.54	30.97	0.0001
COPD	8.34	9.08	8.60	6.19	8.37	0.3448
Epworth	7 (4–9)	6 (4–8)	13 (11–16)	8 (5–11)	10 (6–14)	<0.0001
OSA severity indices						
AHI, events/h	26 (13–41)	23 (13–37)	26 (13–41)	34 (20–52)	27 (14–42)	<0.0001
3% ODI, events/h	20 (10–35)	17 (8–31)	18 (8–35)	26 (12–47)	19 (9–36)	<0.0001
T90	2 (0–8)	1 (0–6)	1 (0–7)	3 (1–12)	2 (0–7)	<0.0001
Hypoxic burden, %min/h	33 (13–70)	26 (11–55)	30 (12–69)	53 (23–118)	32 (13–71)	<0.0001
Polysomnography, %	44.99	53.13	56.65	44.01	52.13	0.0012
CPAP treatment, %	37.61	33.48	43.54	49.47	41.1	<0.0001
MACE, %	12.22	10.31	10.00	14.79	11.05	0.0041

Definition of abbreviations: AHI = apnea-hypopnea index; COPD = chronic obstructive pulmonary disease; CPAP = continuous positive airway pressure; CV = cardiovascular; MACE = major adverse CV events; ODI = oxygen desaturation index; OSA = obstructive sleep apnea; T90 = percentage of recording time with SaO₂ <90%.

Data are expressed as medians (interquartile range) or percentages.

significantly younger than the other subtypes. When compared with other subtypes, the disturbed sleep subtype had the highest proportion of women and the lowest AHI, and the moderately sleepy subtype had the highest age, AHI, and HB.

Predictors of CV Morbidity and All-Cause Mortality

After a median follow-up of 78 months (interquartile range, 52–109), 592 (11.05%) patients experienced the composite outcome including 292 deaths and 300 CV events

(coronary heart disease: 48.9%; strokes: 27.3%, heart failure: 20.6%; simultaneous coronary heart disease and heart failure: 3.2%).

Unadjusted analysis showed that the “excessively sleepy” subtype patients had

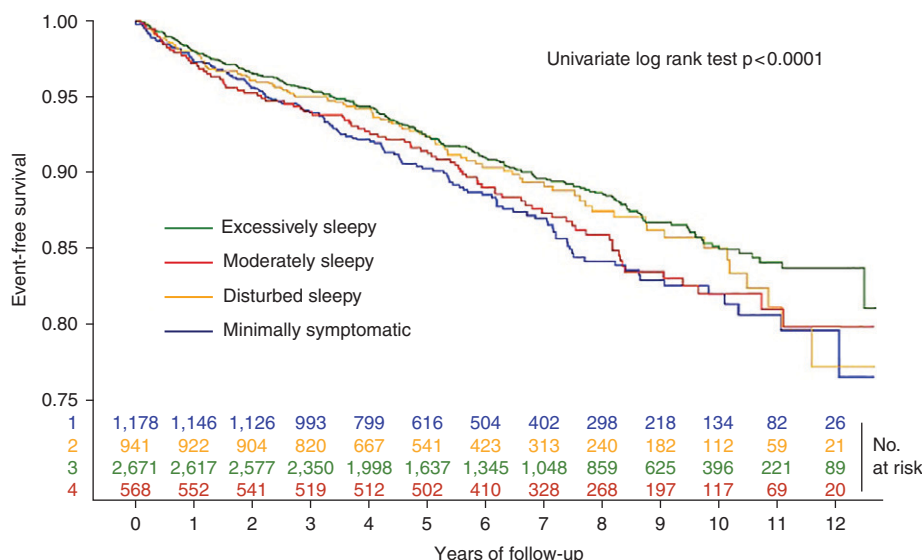


Figure 3. Unadjusted Kaplan-Meier survival curves indicating the time to onset of cardiovascular events and all-cause mortality according to obstructive sleep apnea phenotypes. The log-rank test was used to compare the survival distribution across categories.

Table 2. Cox Proportional Hazards Models Assessing the Association between Incident Cardiovascular Events and All-Cause Mortality and Clinical Subtypes, AHI, Indices of Nocturnal Hypoxemia, and Pulse Rate Variability

	Unadjusted [HR (95% CI)]	P Value	Model 1 [HR (95% CI)]	P Value	Model 2 [HR (95% CI)]	P Value
OSA subtypes						
Minimally symptomatic	1.33 (1.09–1.63)	0.0058	0.99 (0.77–1.27)	0.9459	0.99 (0.77–1.27)	0.9450
Disturbed sleep	1.10 (0.87–1.38)	0.4412	1.13 (0.77–1.65)	0.5429	1.13 (0.77–1.65)	0.5440
Excessively sleepy (ref)	—	—	—	—	—	—
Moderately sleepy	1.24 (0.97–1.59)	0.0825	0.98 (0.81–1.18)	0.8362	0.98 (0.81–1.18)	0.8384
Log_AHI	1.40 (1.26–1.56)	<0.0001	1.12 (0.97–1.29)	0.1122	1.15 (0.98–1.34)	0.0813
Nocturnal hypoxemia indices						
Log_Hypoxic burden	1.27 (1.19–1.36)	<0.0001	1.19 (1.06–1.33)	0.0040	1.21 (1.07–1.38)	0.0024
Log ODI	1.25 (1.15–1.35)	<0.0001	1.16 (0.93–1.44)	0.1941	1.19 (0.94–1.51)	0.1494
Log T90	1.72 (1.50–1.97)	<0.0001	1.32 (1.15–1.52)	<0.0001	1.34 (1.16–1.55)	<0.0001
Pulse rate variability						
Log ratio LF/HF	0.75 (0.65–0.86)	<0.0001	0.93 (0.85–1.02)	0.1216	0.93 (0.85–1.02)	0.1216
Log LF	0.55 (0.43–0.71)	<0.0001	0.77 (0.57–1.03)	0.0744	0.77 (0.57–1.03)	0.0745
Log HF	1.75 (1.24–2.47)	0.0016	1.20 (0.80–1.78)	0.3763	1.20 (0.80–1.78)	0.3764
Log SNND	0.82 (0.68–0.98)	0.0294	0.94 (0.86–1.03)	0.2031	0.94 (0.86–1.03)	0.2034
Log RMSSD	1.05 (0.90–1.23)	0.5009	0.99 (0.92–1.09)	0.9863	0.99 (0.92–1.09)	0.9875

Definition of abbreviations: AHI = apnea-hypopnea index; CI = confidence interval; HF = high frequency; HR = hazard ratio; LF = low frequency; LF/HF = normalized low-frequency power to normalized high-frequency power ratio; ODI = oxygen desaturation index; OSA = obstructive sleep apnea; RMSSD = root mean square of the successive normal-to-normal beat; SDNN = SD of the normal-to-normal beat intervals; T90 = percentage of recording time with $\text{SaO}_2 < 90\%$.

Model 1: adjusted for age, sex, body mass index, smoking status, presence of prevalent disease (diabetes, chronic obstructive pulmonary disease, and hypertension), type of sleep study, study site, and β -blocker/calcium channel blocker medications.

Model 2: model 1 + continuous positive airway pressure treatment.

Data are expressed as median (interquartile range).

the lowest risk of incident MACE as compared with the three other subtypes (Figure 3). Results from unadjusted and adjusted Cox proportional hazards are summarized in Table 2. As expected, AHI, ODI, T90, HB, and most of the natural log transformed (ln) PRV indices (ln SNND, ln LF, ln HF, and ln LF-to-HF ratio) were associated with the occurrence of MACE in the unadjusted model. Significant differences were also observed regarding the association of symptom subtypes and incident MACE, with the minimally symptomatic subtype demonstrating increased risk as compared with other symptom subtypes (hazard ratio [HR], 1.33; 95% confidence interval [CI], 1.09–1.63; $P = 0.0058$ vs. “excessively sleepy” group) (Figure 3). However, only HB and T90 remained significantly associated with MACE after adjusting further for confounders (HR, 1.21; 95% CI, 1.07–1.38; $P = 0.002$ and HR, 1.34; 95% CI, 1.16–1.55; $P < 0.0001$, respectively).

Unadjusted and adjusted Kaplan-Meier survival curves indicating event-free survival according to quartiles of HB are shown in Figure 4. Log-rank tests comparing curves showed that event-free survival was negatively associated with increasing HB ($P < 0.0001$).

The independent association with HB remained significant when all-cause mortality was considered separately (HR, 1.28; 95% CI, 1.06–1.54; $P = 0.01$), but results did not reach statistical significance when only nonfatal CV events were included in the model (HR, 1.11; 95% CI, 0.94–1.30; $P = 0.21$). Adjusted Kaplan-Meier survival curves indicating the time to incidence of all-cause mortality and nonfatal CV events grouped by quartiles of HB are shown in Figure 5.

Stratified analyses suggested a stronger association of high HB with MACE in women than in men (P for interaction = 0.03) and in younger patients (≤ 65 yr; P for interaction = 0.057). The association was also stronger in patients recorded using HSAT as compared with PSG. No further differences were observed when patients were stratified according to BMI. In unadjusted analysis, there was a significant correlation between CPAP use and MACE. However, in the fully adjusted model, CPAP treatment had no significant impact on the major outcome (HR, 0.91; 95% CI, 0.76–1.09; $P = 0.31$). Adjusted Kaplan-Meier survival curves indicating event-free survival according to quartiles of HB in men, women, and younger and older patients are shown in Figure 6.

Discussion

This study provides further evidence of the existence of clinical symptom subtypes of OSA described within a number of previous studies, encompassing both clinical and population-based samples (16). However, in contrast to recently published data based on population-based samples, we found that those subtypes cannot be considered as independent CV prognostic tools in a clinical-based sample. Conversely, data regarding the independent prognostic value of HB to predict incident CV events and deaths, previously described in population-based samples, are confirmed in our large multicenter clinical-based cohort.

To date, several studies provide convincing evidence that similar symptom-based subtypes of OSA are found within patients identified either from sleep clinics or in the general population (16). Similar to recent analyses in the Sleep Apnea Global Interdisciplinary Consortium (SAGE) cohort (14) and the SHHS (18), the current study found four distinct subgroups when we looked at the optimal cluster solution. Although similar symptom subtypes were found, significant differences must be acknowledged regarding the symptoms included in the latent class analysis. In the study of Mazzotti and

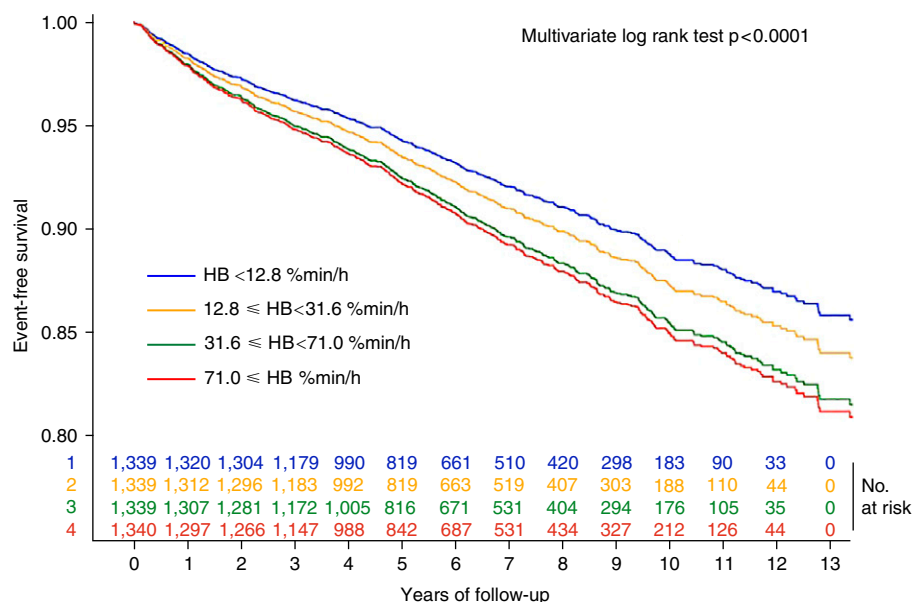


Figure 4. Adjusted Kaplan-Meier survival curves indicating the time to onset of cardiovascular events and all-cause mortality in the whole population. The log-rank test was used to compare the survival distribution across categories. Adjustment included age, sex, body mass index, smoking status, presence of prevalent disease (diabetes, chronic obstructive pulmonary disease, and hypertension), type of sleep study, study site, β -blocker/calcium channel blocker medications, and continuous positive airway pressure treatment. HB = hypoxic burden.

colleagues, latent class analysis was based on 15 different symptoms as compared with the 7 symptoms included in the present study. As expected, more pronounced differences were observed regarding subtype frequency across samples between clinical-based and population-based cohorts. The prevalence of the “excessively sleepy” subtype was evaluated between 17% and 30% in population based-

cohorts (17, 18). In contrast, almost half of our sample was classified in this subtype, which is in accordance with the prevalence observed in clinical samples (13, 14). Consequently, a lower percentage of asymptomatic or moderately symptomatic patients was found in our cohort (32.6%) than in population-based cohort from the SHHS (71.1%) (18).

Our clinical cohort study did not confirm previous findings from community-based cohorts that have suggested an independent link between OSA severity and CV morbidity and mortality mainly in male and “excessively sleepy” subjects (4–6, 18, 26). The distinct way of patients’ inclusion in population-based and clinical cohorts might explain the opposite results that were found

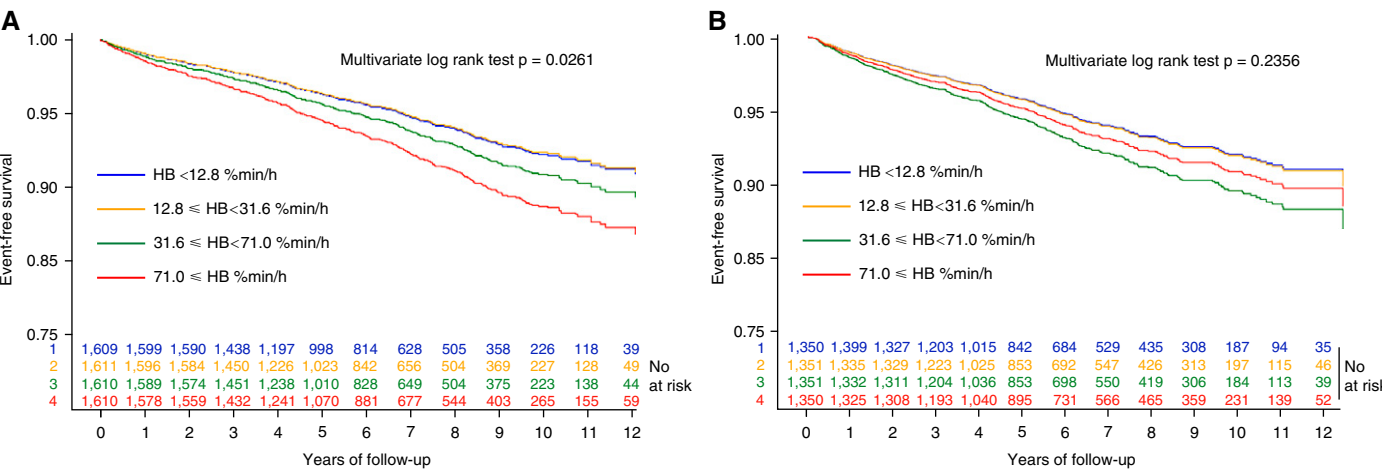


Figure 5. (A and B) Adjusted Kaplan-Meier survival curves indicating the time to onset of cardiovascular events (A) and all-cause mortality (B). The log-rank test was used to compare the survival distribution across categories. Adjustment included age, sex, body mass index, smoking status, presence of prevalent disease (diabetes, chronic obstructive pulmonary disease, and hypertension), type of sleep study, study site, β -blocker/calcium channel blocker medications, and continuous positive airway pressure treatment. HB = hypoxic burden.

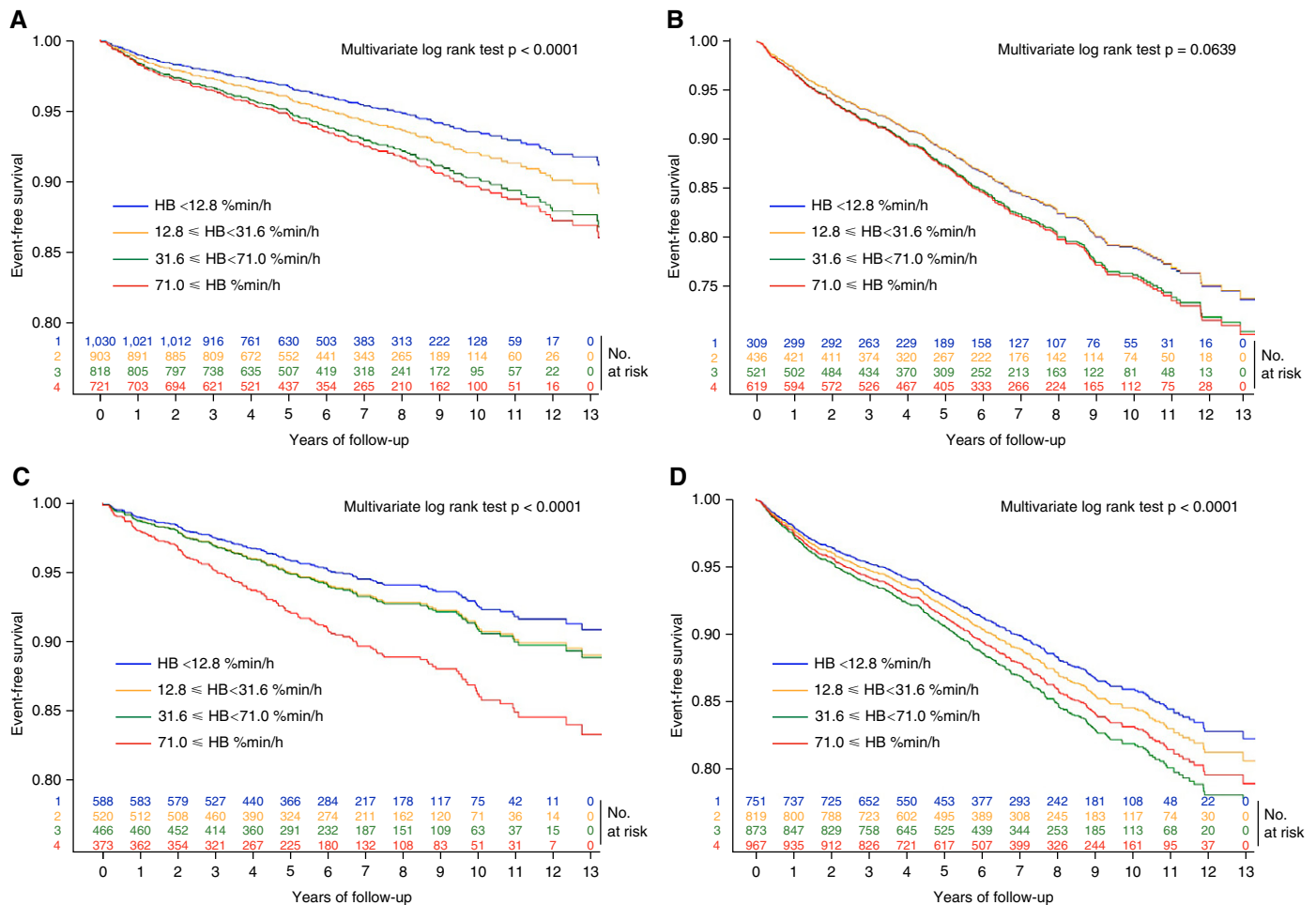


Figure 6. (A–D) Adjusted Kaplan-Meier survival curves indicating the time to onset of cardiovascular events and all-cause mortality in patients ≤ 65 years of age (A), patients > 65 years of age (B), men (C), and women (D). The log-rank test was used to compare the survival distribution across categories. Adjustment included age, sex, body mass index, smoking status, presence of prevalent disease (diabetes, chronic obstructive pulmonary disease, and hypertension), type of sleep study, study site, β -blocker/calcium channel blocker medications, and continuous positive airway pressure treatment. HB = hypoxic burden.

in our sample as compared with the SHHS regarding the prognostic value of the different clinical subtypes. The study by Mazzotti and colleagues was a community-based cohort study of participants aged > 40 years with an AHI ≥ 15 /h. In contrast, we included patients attending sleep centers with an AHI ≥ 5 /h and no overt CV disease. Consequently, patients from our study were slightly younger, with a lower AHI and a lower proportion of prevalent comorbidities, and those differences might contribute to explaining the discrepancies in findings. Mazzotti and colleagues reported that the “excessively sleepy” subtype was associated with the poorest CV prognosis in unadjusted and fully adjusted analysis. Those data were in accordance with previous studies

reporting that sleepiness could represent *per se* a risk factor for hypertension (27) and incident CV events (28, 29). However, this observation is not homogeneous across all the studies, and no combined effect of OSA and excessive daytime sleepiness was recently found on incident CV events in the SHHS (30). In contrast, we found that in the unadjusted analysis, the “minimally symptomatic” subtype was associated with the poorest prognosis and this difference was significant as compared with the “excessively sleepy” subtype. We hypothesized that patients from the “minimally symptomatic” subtype were likely referred to the sleep clinics due to OSA-related comorbidities rather than symptoms and that these comorbidities may have impacted CV risk.

Indeed, the prevalence of hypertension and diabetes is the highest in this subgroup as compared with the three others. After further adjustments for these covariates, the association was no longer significant. These findings suggest that data from population-based cohorts regarding the predictive value of symptom subtypes should not be extrapolated to clinical practice.

We found that HB and T90 were better associated with MACE than AHI and ODI. Recent evidence suggests that neither AHI, nor ODI, nor T90% might represent the accurate OSA severity marker to predict major OSA comorbidities. AHI and ODI might fail to capture the keys aspects of OSA that have a deleterious impact on CV prognosis. Indeed, they provide a simple

count of the number of obstructive episodes or oxygen desaturations per hour of sleep, without regard for the duration and the depth of respiratory disturbance or blood gas changes. In line with recent reports (19, 20), our findings suggest that compared with common metrics of OSA severity such as AHI and ODI, HB might better characterize OSA-related nocturnal desaturations and predict MACE incidence in a clinical setting. As previously observed (7), we also found an association between MACE and the extent of overall nocturnal hypoxemia assessed by T90. Unlike HB, T90 is provided in all sleep analysis software. However, the T90 characterizes not only obstructive events related intermittent hypoxemia, but also sustained hypoxemia that might be consecutive to various respiratory conditions such as chronic obstructive pulmonary disease or obesity hypoventilation syndrome. As pulmonary function tests were not available in our database, a contribution of underlying comorbid respiratory disorders to the association of T90 with MACE cannot be excluded.

The association between MACE and HB remained significant with all-cause mortality but not with incident nonfatal CV events when the two outcomes were considered separately. The lower rate of events considered in this complementary analysis might participate to the absence of significant association. Alternatively, one can hypothesize that sleep apnea-associated hypoxemia might be more strongly involved in the fatal outcome of CV events than in the development of incident nonfatal CV comorbidities. This hypothesis is supported by the high percentage of nocturnal death from cardiac causes in the OSA population as compared with subjects without OSA (31) and by a recent meta-analysis showing that OSA is a risk factor for all-cause mortality, cardiac mortality, and sudden death (32). We consider that MACE represents a robust and relevant clinical outcome regularly assessed in randomized controlled trials and prospective cohorts (33–35). Our data confirm those from previously published population-based samples that HB is a strong predictor of incident CV morbidity and mortality that could represent a useful tool in clinical practice (19, 20, 22).

Stratified analysis showed a stronger association between HB and MACE in women and younger patients. To date, the current available data on OSA and mortality do not provide a definitive

answer as to whether sex differences in susceptibility exist (8). Both clinical and population-based cohorts have suggested a stronger association among women (35–37). However, this observation was not confirmed in the SHHS (5, 6, 26), but the number of recruited female patients with severe OSA in the SHHS was particularly small (around 3%). A clearer consensus exists in both clinical and population-based cohorts regarding a stronger CV impact of OSA in middle-aged as compared with older patients (5, 38–41). In unadjusted analysis, we found a significant correlation between CPAP use and MACE. As shown in a previous study from the same cohort (24), patients using CPAP are generally more severe than patients who do not require or are not adherent to treatment. Those patients are older, with a higher BMI and frequent comorbidities that might explain the observed higher CV risk. In the fully adjusted model, CPAP treatment had no significant impact on the major outcome. Of note, our study was not designed and statistically powered to evaluate any protective effect of CPAP therapy against MACE. Finally, the association between HB and MACE was stronger in patients investigated using HSAT as compared with PSG. According to French guidelines, HSAT is the first-line diagnostic tool proposed to patients with a high clinical probability of OSA. Consequently, patients recorded using HSAT were older, with a higher BMI and higher frequency of comorbidities (Table E2) that might explain the differences observed when compared with the PSG-recorded patients.

We also evaluated different markers of PRV on the CV prognosis and failed to find any independent association. Recent publications based on the same cohort found that PRV was an independent predictor of incident AF and stroke (22, 23). PRV reflects the autonomic nervous system activation associated with apnea and hypopneas, and this activity was demonstrated to play a specific critical role in the initiation and maintenance of AF (42). It is therefore plausible that PRV is a stronger predictor of incident AF and stroke, its major complication, but does not represent the best overall CV risk predictor in our clinical sample.

The strengths of the current study include a multicenter design and a wide

range of sleep-disordered breathing severity, suggesting that our findings are generalizable to most patients investigated for OSA. In addition, our study includes adjustments for multiple CV risk factors, a high percentage of patients matched to administrative data, a long and complete follow-up with access to comprehensive SNDS data, and the assessment of clinical parameters and different measures of hypoxia and PRV. The ability to study clinical outcomes in large population- or clinic-based samples is a potential strength of health administrative data. The linkage between clinical cohort data and SNDS data makes easier the long-term follow-up with limited attrition bias as compared with trials (43, 44). In French hospitals, all physicians routinely contribute to SNDS data collection, with annual quality control of coding resulting in a decreased risk of misclassification bias (45).

Our study also has several limitations. The algorithms used in the study assessed all-cause mortality and did not clarify the proportion of deaths specifically attributable to CV events. Furthermore, no adjudication of the primary outcome was performed. As detailed, MACE was identified using an algorithm based on a French healthcare database. This database covers 98.8% of the French population, >66 million people, from birth (or immigration) to death (or emigration) and contains comprehensive data on all reimbursements for health-related expenditure, as well as demographic data and medical information on all admissions to French public and private hospitals (according to the International Statistical Classification of Diseases, 10th Revision). We acknowledge that the algorithm used for identifying MACE might have resulted in misclassifications. However, recent studies found convincing performances of administrative databases in identifying major CV outcomes and deaths (46).

The CPAP usage was closely followed, which is a strength of this study. In contrast, alternative treatments such as mandibular advancement device (MAD) use were not recorded in this study. Owing to the limited use of MAD for OSA treatment, its lower efficacy in reducing AHI compared with CPAP, and the lack of objective measure of MAD compliance, we considered that these patients did not receive any active OSA therapy in our analysis.

We also acknowledge that sleep recordings were performed by either PSG or HSAT and manually scored

locally, with no assessment of interscorer reliability. It cannot be excluded that interscorer variability, particularly for hypopneas that did not cause an oxygen desaturation, may have contributed to the lack of association between MACE risk and OSA severity as assessed by AHI (47). However, this potential bias

was mitigated by the adjustment for study site.

Conclusions

In patients investigated for clinical suspicion of OSA, HB was an independent predictor of incident CV events and death. HB may be used, in clinical practice, to identify patients

with OSA at greatest risk of CV events. In contrast to observations from population-based cohorts, symptom subtypes and PRV have no predictive value in clinical-based samples. ■

Author disclosures are available with the text of this article at www.atsjournals.org.

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